

Problem-Solution Fit Canvas

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

Canvas Dimension	Observed Customer State / Problem	Validated Fit Solution (AI-FitTrack-API)
1. Customer Segment [CS]	Fitness enthusiasts, busy professionals, and home workout athletes seeking structured guidance without expensive personal trainers.	Developer-ready REST API providing instantaneous, tailored workout generation and activity persistence for web/mobile apps.
2. Problems / Pains [PR]	Generic internet routines cause rapid plateaus; personal trainers cost \$100+/hr; paper diaries and fragmented apps lose history.	Google Gemini 1.5 Flash integration crafts adaptive, equipment-specific plans; centralized MongoDB logs ensure historical continuity.
3. Available Solutions [AS]	Commercial apps (Strava, MyFitnessPal) charge recurring subscriptions and hide analytics behind paywalls; notes apps lack structure.	100% open, free REST API with stateless JWT authentication, automated calorie calculations, and transparent AI prompt responses.
4. Customer Limitations [CL]	Limited home equipment (e.g. only dumbbells), constrained daily workout duration (20-45 mins), and variable daily energy levels.	Prompt inputs accept availableEquipment and duration parameters, generating customized routines matching constraints exactly.
5. Solution Capabilities [SC]	Automated workout logging across 4 categories, MET calorie estimations, and AI insights evaluating consistency and recovery.	Sub-50ms CRUD latency, structured JSON routine outputs, and multi-factor consistency milestones preventing overtraining.